Memo

2450 Venture Oaks Way, Suite 500, Sacramento, California 95833 (916) 648-9700 - FAX (916) 648-6534

Date: 5 June 2023

To: Maria Zupo
Kitchell CEM, Inc., Project Manager II

From: Mohd. Sharif Mohammadi
Kitchell CEM, Inc., Scheduling Services

Regarding: Merritt College New Landscape Horticulture Complex – Baseline 03 Schedule Review

Purpose of Memo

Kitchell CEM, Inc. ("Kitchell") has prepared a technical analysis of the baseline schedule. The file is titled "446 - MERRITT-BL-1.xer", received by Kitchell on 30 May 2023. This memo is to communicate the observations from the review of the schedule file. The schedule was reviewed for compliance with contract requirements and general scheduling best practices.

Schedule Disposition

☐ Revise and Resubmit
☒ Accepted

Based on this review, the current schedule is recommended as accepted.

Executive Summary

Milestone Overview

The Notice to Proceed (NTP) for Merritt College New Landscape Horticulture Complex is February 06, 2023, with 540 calendar days to complete on July 29, 2024. Schedule variance is shown below.

Contractual Milestones/ Variance Tracking						
Activity ID	**Contractual Milestones**	**Contract**	**Baseline 01**	**Baseline 02**	**Baseline 03**	**Variance**
PM 1000	NTP	Feb-06-23	Feb-06-23	Feb-06-23	Feb-06-23	0
PM 1030	All Submittals Provided	May-07-23	May-07-23	Feb-27-23	Feb-27-23	69
PM 1040	Slab on Grade Complete	Jul-13-23	Jul-13-23	Jul-13-23	Jul-13-23	0
PM 1050	Building Weather Tight	Feb-02-24	Nov-03-23	Feb-02-24	Feb-02-24	0
PM 1060	Substantial Completion	Jun-30-24	Aug-20-24	Jun-17-24	Jun-18-24	12
PM 1010	Final Project Completion	Jul-29-24	Sep-30-24	Jul-29-24	Jul-29-24	0

Project Budget

The project budget is outlined below. The following table shows the original contract plus amendments to date, completed work, remaining work, and percent expended.

Original Contract	Change Order	Current Contract	Completed Work	Remaining Work	% Budget	% Duration
$24,712,615	$0	$24,712,615	$0	$24,712,615	0%	0%

Project Planned Value

Schedule Performance

The graph illustrates the contractor's work plan cash flow distributed in a 'S'-Curve over the life of the project. As the project progresses, each month earned value metrics can be applied using the planned curve

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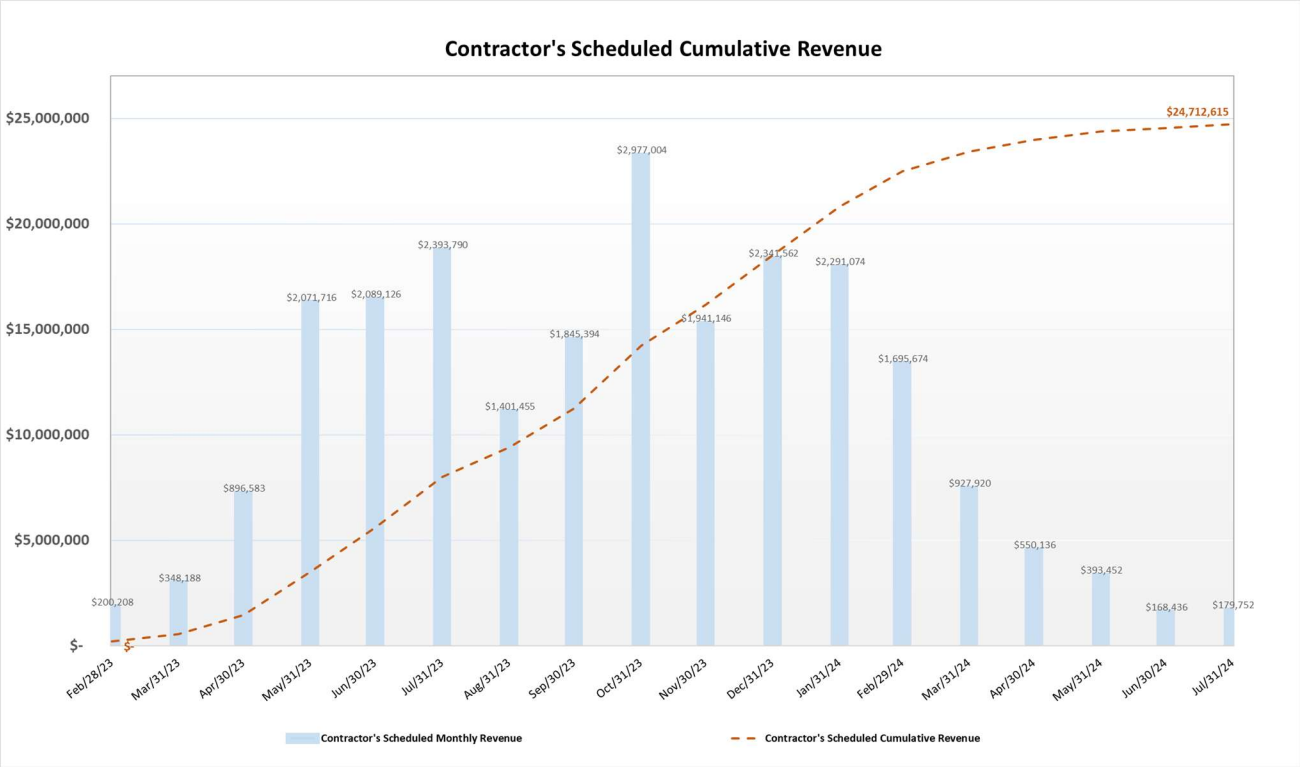
and the payment application. This is for the purpose of assessing schedule performance and to forecast a reasonable completion date, if necessary. This methodology is crucial to understanding progress aside from milestone variance and activity execution.



Constructability Analysis

Duration Probability

The construction activity durations have been assessed below based on original (optimistic), most likely, and pessimistic scenarios to arrive at a probable duration. This process calculates the mean distribution or expected duration and the number of standard deviations from the mean to reach a likely duration. The results show that **based on the sum of the baseline's critical activity durations**, the percent probability the project will complete as planned.

Critical Path			
Optimistic	Most Likely	Pessimistic	MEAN
967	1,038	1,118	1,040
% Probability	Critical Path Std Dev (Sigma)		
50% (MEAN)	1σ	3.67	1,039.58
68%	2σ	7.33	1,046.91
95%	3σ	11.00	1,050.58

Probable Critical Activity Summation Range		
99.7%	11.00	1,051
99.7%	(11.00)	1,029
Equivalent Project Duration		
Activities- Phase 2	Optimistic Summary (Critical Path)	Optimistic Duration (Project)
Baseline	967.00	540
High range	1,050.58	587
Low Range	1,028.58	574

Contract Duration

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- Based on the baseline activity durations, there is a 99.7% probability that the project completion will finish within 574 and 587-days (6%-9% longer than current contractual duration).

Critical Analysis

Longest Critical Path

The critical path is the longest sequence of activities which must be completed on time for the project to complete as scheduled. It includes the minimum time necessary to complete the entire project from start to finish.

The longest critical path begins with NTP, Mobilization/Site Preparation and Abatement/Demolition including Video Documentation, Remove and Salvage Items, Power/Water, Office Trailer/Container/Laydown Area, Temp, Set-up Containment & Abate Existing Buildings, Safe-off & Air Gap Verify Building Utilities, Demo of Existing Buildings, Clear & Grub, Remove Existing Site Concrete & Asphalt, Remove Existing Underground Utilities, Survey Building & Retaining Wall Locations scheduled to complete on **3/20/23**.

Next is Site Work and Foundation & Slab activities including Rough Grade, Cut, & Fill, Install Underground Utilities, Fine Grade, Establish Building & Footing Lines for Greenhouse/Soil Bins/Headhouse, Place Footing Concrete, Edge Form & Column Blockouts, Gravel & Vapor Barrier, Slab & Curb Rebar, Place & Finish Slab / Curbs ending on **6/23/23.**

Next, Exterior Wall Framing for Class Lab1/Lab2/Lab3/Library ending on **10/19/23**.

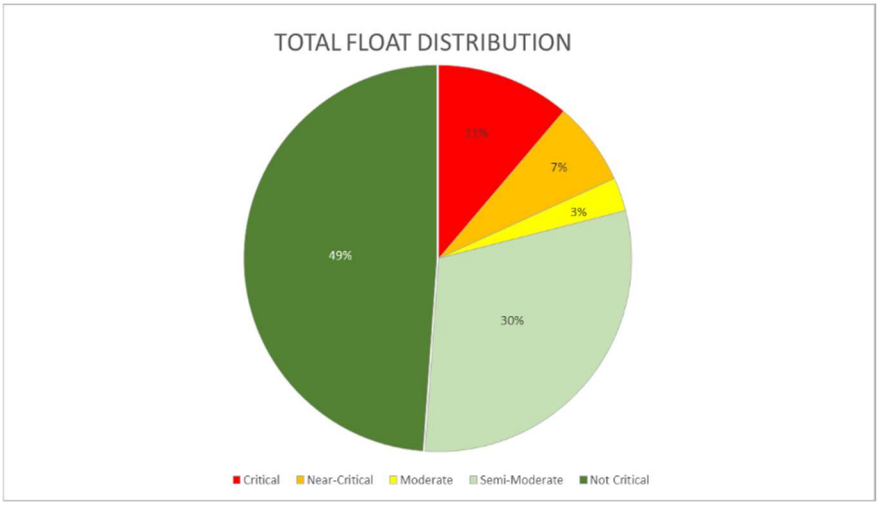
Next is Library Interior Wall Framing, Roof Framing, Sheathing, Roof Sheathing, Soffit / Ceiling Framing, Hang Drywall, Tape & Texture, Paint, Acoustical Ceiling Tiles, Polish Floors, Tackboard & Wainscot, MEP Rough-In, Exposed Ductwork, Sprinklers, MEP Trim scheduled to complete on **5/29/24.**

Finally, Closeout activities including Punch List Development, Contractor Correction of Punch List, Final Cleanup, Demobilization, Final Inspection, and Notice of Completion on **7/29/24.**

- See attached Longest Critical Path report for more detail.

Total Float Distribution

Total float is the amount of time an activity can be delayed without delaying the overall project duration. This exercise is to explain total float distribution and different levels of project delay risk. The following chart depicts the distribution of float among all scheduled activities:



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Each month the total float distribution will be compared to, along with the earned value analysis, to understand project progress and health. Of the overall 571 activities, distribution of activities is as follows:

1. 11.2% of the baseline activities are critical.
2. 7.0% of the baseline activities are near critical.
3. 18.2% of the baseline combined critical and near critical activities.

- Contract schedule specifications require limiting this to no more than 25% combined. The Critical/Near Critical percentages in the current schedule meet the contract schedule requirements.

Technical Analysis and Contract Compliance

The following do not represent scheduling best practices and should be considered for revision.

Logic Analysis

Best Practice	Occurrences	Issue / Specification	Action
Source		**Logic Analysis**	
Activities with Finish-Start + Lag relationships	1	Can hide float	Ok
Activities with Start-Finish relationships	0	Not widely accepted as standard	Ok
Activities with Open Ends	0	Can skew total float and critical path NTP and Final Completion are acceptable	OK
Out-of-Sequence	0	Can skew critical path	Ok
Negative Lag	0	Can be better represented with alternate relationship	Ok

- See the attachments to view the complete Logical Analysis Report.

Conclusion and Action Items

- 18.2% of the baseline combined critical and near critical activities.
- The weather allowance should be critical and follow all work (Last construction activity) while preceding substantial completion. This has been discussed – no action.

Attachments

1. All Activities
2. Longest Critical Path
3. Near Critical Path – Multiple Float Paths
4. Logic Analysis

Disclaimer

The findings, observations, and/or opinions expressed in this review are not intended to direct the Contractor to revise its intended methodology or act as written authorization to amend any part of any of the Contract documents. Any questions or comments regarding this schedule review can be directed to Kitchell's project manager. Additionally, the review and rejection or non-rejection of any schedule, or any part of a schedule shall not be construed as a complete review and analysis of the submitted schedule(s). The acceptance of a schedule or not rejecting any part of it does not relieve the Contractor from any requirements in the Contract, nor is the acceptance an authorization to change any terms or conditions of the underlying Contract Specifications without the Owner's written authorization. The Contractor is respectfully reminded that failure of the Contractor to include any element of the Work within the schedule or any inaccuracy in the schedule will not relieve responsibility for accomplishing the Work in accordance with the Contract documents and within the Contract time. The Contractor is solely responsible for ensuring that all subcontractors and suppliers comply with the requirements of the Contract schedule for their portions of the Work, and that all schedules submitted to the Owner have been developed after coordination with subcontractors and suppliers. If the Contractor believes they are entitled to a Contract time extension, they are respectfully reminded that they are obligated to prepare and submit a time extension request that is consistent with Contract requirements to maintain and established entitlement. Failure to provide the necessary items and documentation may result in the automatic waiver by the Contractor for an extension and may also result in the Contractor in having to recovery or mitigate any loss time at its own cost.

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